Introduction to Probability and Statistics

Lecture 12: Confidence Intervals (Cont.)

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1. Confidence Intervals

2. Tests of Hypothesis

First strategy: pivotal function

Example: C.I for the parameters of a normal distribution

Theorem

Let $X_1, \cdots, X_n$ be n i.i.d. r.v.'s from $\mathcal{N}(\mu, \sigma^2)$.

(i) $\overline{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$.

(ii) $(n-1)\widehat{\sigma}_n^2/\sigma^2 \sim \chi^2_{n-1}$.

(iii) $\overline{X}_n$ *and* $(n-1)\widehat{\sigma}_n^2/\sigma^2$ *are independent r.v.'s.*

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  where α_1 and α_2 are such that $\alpha_1 + \alpha_2 = \alpha$.
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  $$a = \overline{X}_n - \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1 - \alpha_1) \;\; and \;\; b = \overline{X}_n - \frac{\sigma}{\sqrt{n}}\Phi^{-1}(\alpha_2),$$

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  where Φ is the inverse function of the c.d.f of a standard normal distribution.
- If $\alpha_1 = \alpha_2 = \alpha/2$, the interval is symetric and the bounds are:
  $$\boxed{\overline{X}_n \pm \frac{\sigma}{\sqrt{n}}\Phi^{-1}(1-\alpha/2)}$$

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$$a = \frac{(n-1)\ \widehat{\sigma}_n^2}{\chi_{n-1}^{2^{-1}}(1-\alpha_1)} \quad \textit{and} \quad b = \frac{(n-1)\ \widehat{\sigma}_n^2}{\chi_{n-1}^{2^{-1}}(\alpha_2)},$$

where $\chi_{n-1}^{2^{-1}}$ is the inverse function of the c.d.f of a chi-squared distribution with $(n-1)$ d.o.f.

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where t^{-1}_{n-1} is the inverse of the c.d.f. of a Student r.v. with $n-1$ d.o.f.

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where t_{n-1}^{-1} is the inverse of the c.d.f. of a Student r.v. with $n-1$ d.o.f.

➥ If $\alpha_1 = \alpha_2 = \alpha/2$, $\boxed{\overline{X}_n \pm \dfrac{\widehat{\sigma}_n}{\sqrt{n}}\ t_{n-1}^{-1}(1-\alpha/2)}$ since $t_{n-1}^{-1}(\nu) = -t_{n-1}^{-1}(1-\nu)$.

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Example : Confidence interval for the parameter p of a Bernoulli r.v.
Let $(X_1, \cdots, X_n)$ be n independent Bernoulli r.v.'s with parameter p.
Recall that: $E(X_i) = p$ et $Var(X_i) = p(1-p)$, $i = 1, \cdots, n$.
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Applying the CLT, we have:

$$Z_n = \frac{\overline{X}_n - p}{\sqrt{p(1-p)/n}} \sim \mathcal{N}(0,1).$$

Slusky's Theorem

Let $X_1, \cdots, X_n$ be n r.v.'s and let $\varphi_1(\underline{X}), \ldots, \varphi_r(\underline{X})$ be r functions of the r.v.'s. Suppose that $\varphi_1(\underline{X}) \xrightarrow{P} a_1, \ldots, \varphi_r(\underline{X}) \xrightarrow{P} a_r$ (as $n \to +\infty$) for some constants $a_1, \ldots, a_r$. Suppose that $\psi(., \ldots, .)$ is a continuous function such that $|\psi(a_1, \ldots, a_r)| < +\infty$. Then:

$$\psi(\varphi_1(\underline{X}), \ldots, \varphi_r(\underline{X})) \xrightarrow{P} \psi(a_1, \ldots, a_r).$$

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$$P(a < p \leqslant b) = 1-\alpha \Longleftrightarrow P\left(\frac{\overline{X}_n - b}{\sqrt{\widehat{p}(1-\widehat{p})/n}} < Z_n \leqslant \frac{\overline{X}_n - a}{\sqrt{\widehat{p}(1-\widehat{p})/n}}\right) = 1-\alpha_1-\alpha_2,$$

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where α_1 and α_2 are such that $\alpha_1 + \alpha_2 = \alpha$.

- The bound of the CI are then given by:

$$a = \overline{X}_n - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}}\Phi^{-1}(1-\alpha_1) \quad and \quad b = \overline{X}_n - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}}\Phi^{-1}(\alpha_2).$$

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Example of computation

```
n<-100
p<-0.3
set.seed(23)# to reproduce the same set of random numbers.
x<-sample(c(0,1), n,replace=TRUE, c(1-p,p));x
#or
x<-rbinom(n,1,p);x
hp<-mean(x);hp
alpha1<-alpha2<-0.05
a<-hp-sqrt(hp*(1-hp)/n)*qnorm(1-alpha1);a
b<-hp-sqrt(hp*(1-hp)/n)*qnorm(alpha2);b
```

Conclusion: the interval [0.29, 0.45] contains the true value of p with a probability 0.9.

Introduction

- *Point estimation*

  ➥ **Goal:** suggest a value for θ.

$$\hat{\theta} = g_n(X_1, \cdots, X_n) \approx \theta.$$

- *Confidence Interval*

  ➥ **Goal:** find an interval that contain θ with a given probability.

$$[a, b] \ni \theta.$$

- *Test of hypothesis* also called *test of significance*

  ➥ **Goal:** assess the evidence provided by the data in favor of some claim about θ.

$$H_0 : \theta = \theta_0.$$

Vocabulary

- **Null hypothesis** H_0
  - The statement being tested is called the **null hypothesis**.
  - A test of significance is designed to assess the strength of the evidence against the null hypothesis.
  - Usually the null hypothesis is a statement of "*no difference*" or "*no effect*".

Vocabulary

- **Alternative hypothesis H_a**
  - The alternative hypothesis is the statement we suspect to be true instead of H_0
  - We say that H_0 is tested against H_a.
  - Nature of H_a:
    - Two-sided test: $H_0: \ \theta = \theta_0$ versus $H_a: \ \theta \neq \theta_0$.
    - One-sided test: $H_0: \ \theta = \theta_0$ versus $H_a: \ \theta < \theta_0$ or $H_a: \ \theta > \theta_0$.

Decision

A test of hypothesis is a problem of Decision.

After observing $X \in \mathscr{X}$, we "decide" to reject H_0 or to do not reject H_0 (which does not mean that we "accept" H_a).

Therefore $\mathscr{X}$ is split into two regions

- R: the **critical region** or **region of rejection**, if $X \in R$, we reject H_0.
- $\overline{R}$: the region of non-rejection, if $X \in \overline{R}$, we do not reject H_0.

A test of hypothesis can be seen as a *decision*:

$$\delta(x) = \begin{cases} 1, & \textit{if } x \in R, \\ 0, & \textit{if } x \in \overline{R}. \end{cases}$$

Two types of error

- **Error of type I**: rejecting H_0 when it is true.
  - Probability: α
    - **level of significance** of the test or size of the critical region.

$$\alpha = P\left[\delta(X) = 1 \mid \theta = \theta_0\right].$$

- **Error of type II**: do not reject H_0 when it is false.
  - Probability: β
    - $1 - \beta$ is called the **power** of the test.

$$\beta = 1 - P\left[\delta(X) = 1 \mid \theta \neq \theta_0\right].$$

Minimize the type I error / Maximize the power of the test.

- The four steps to build a test of hypothesis:
  - Choose the level of significance α
  - Determine the test statistics
  - Describe the critical region
  - Conclude/ make a decision
    - ➥ **reject** H_0 , the null hypothesis with a probability α to be wrong.
    - ➥ or **do not reject** H_0 *(which does not mean in general that we accept H_0 that is H_0 is true: the experiment does not allow to consider that H_0 is not true.*
      In a court trial, when the members of the jury return a verdict of "No guilty", it does not mean that the defendant is innocent. It simply means that the evidence is not sufficient to overturn the presumption that the defendant is innocent.)

- The probability, assuming H_0 is true, that the test statistic would take a value as extreme or more extreme that actually observed is called the p-**value** of the test.

- The smaller the p-value, the stronger the evidence against H_0 provided by the data.